The Correlation Length of Kinetic Helicity Needed to Drive A Large-scale Dynamo

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“Mean-field dynamos require non-zero mean kinetic helicity.”

… or do they?

1

Hey Vsauce, Barshan here. What *IS* a mean?

The Ergodic Principle

- The ensemble average.
- The temporal average.
- The spatial average.

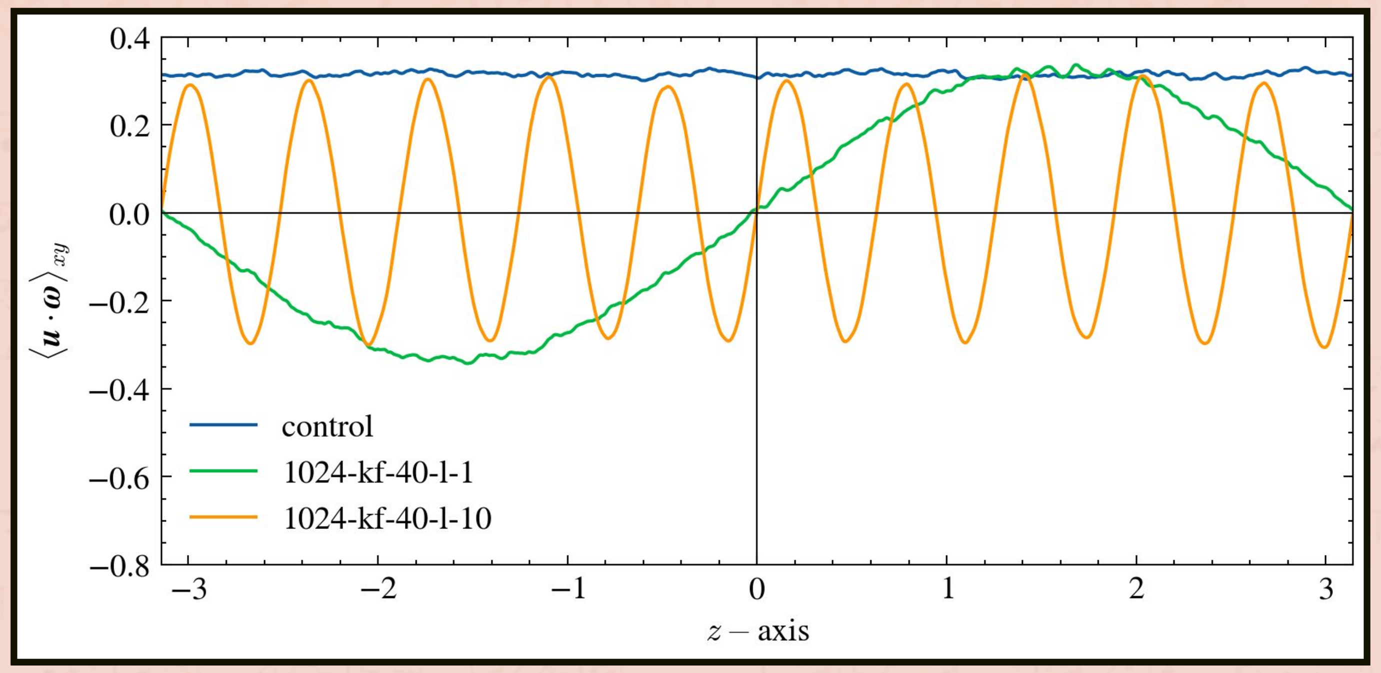


Figure: Mean kinetic helicity variation over the z-axis of three 1024^3 finite grid MHD simulations using the PENCIL CODE.

If one is free to choose the scale over which they define the spatial average, a zero mean over larger scales may be consistent with non-zero means defined over smaller scales.

What if a system consists of lots of local mean-field dynamos operating, leading to a global mean-field dynamo?

P.S. Dynamo = exponential growth of magnetic energy

2

… but what is kinetic helicity?

Scan QR for accompanying material, including:

1. DOI of referred book (Astrophysical Magnetic Fields, Shukurov & Subramanian)
2. Newspaper clippings describing the concerning condition of science in India; these are the backgrounds used for panels 1 & 3.

SCAN ME

$\boldsymbol{u}\cdot\boldsymbol{\omega}$

It is a good proxy for the expected dynamo growth rate, as

$\gamma \propto \sqrt{|\alpha|} \simeq \sqrt{\langle \text{helicity} \rangle}$

BTW, we control the helicity forced with the following forcing function:

$$\boldsymbol{f_k} = \sqrt{\frac{1+\sin lz}{2}}\boldsymbol{f_k}^{(+)}(t) + \sqrt{\frac{1-\sin lz}{2}}\boldsymbol{f_k}^{(-)}(t)$$

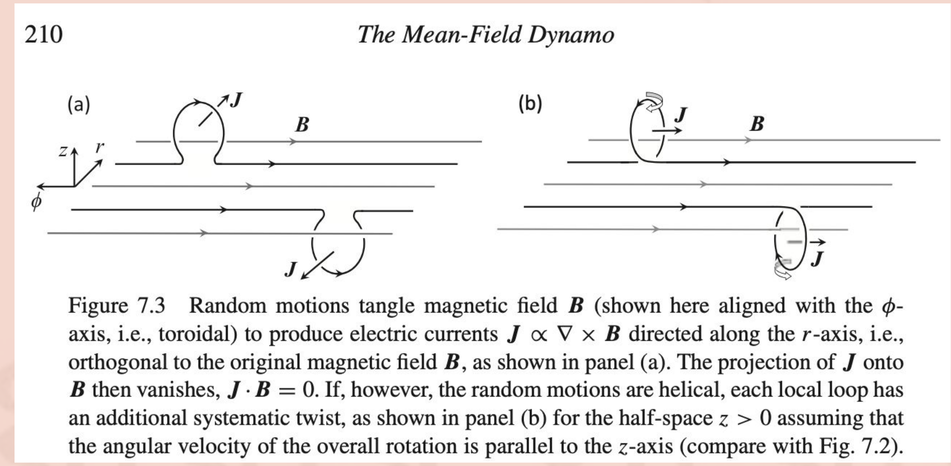
For a fluid to have non-zero kinetic helicity, the fluid velocity $\boldsymbol{u}$ and the vorticity $\boldsymbol{\omega}$ have to have a non-zero dot product.

NO (T REALLY) !

A dynamo still operates, with a fluctuating $\alpha - \Omega$ effect.

3

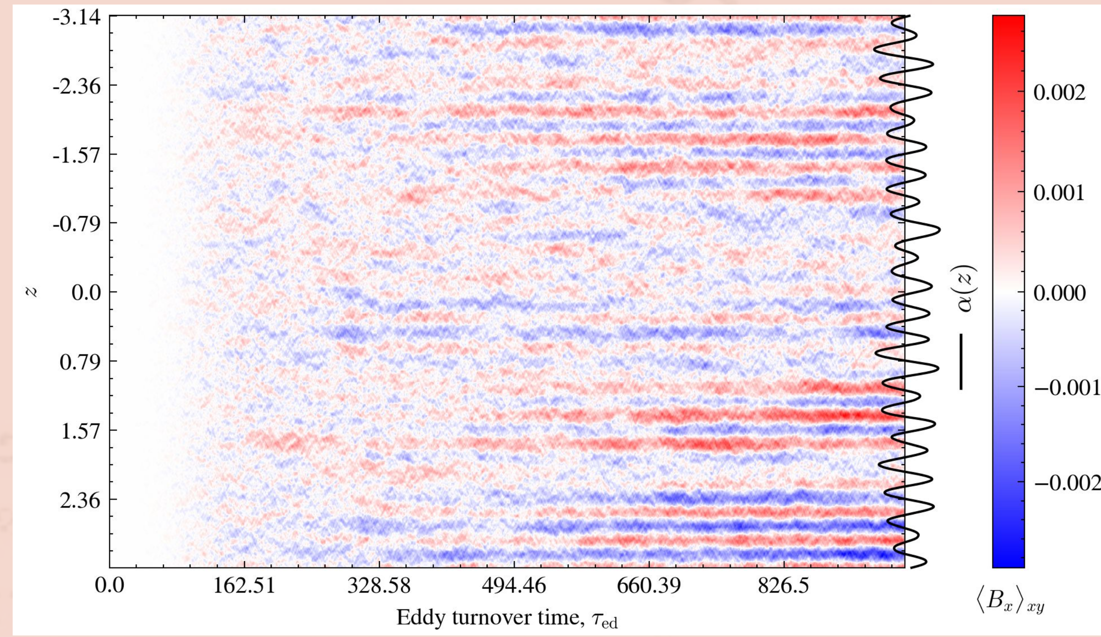
… but why kinetic helicity?


210 *The Mean-Field Dynamo*

(a) (b)

Figure 7.3 Random motions tangle magnetic field $\boldsymbol{B}$ (shown here aligned with the ϕ-axis, i.e., toroidal) to produce electric currents $\boldsymbol{J} \propto \nabla \times \boldsymbol{B}$ directed along the r-axis, i.e., orthogonal to the original magnetic field $\boldsymbol{B}$, as shown in panel (a). The projection of $\boldsymbol{J}$ onto $\boldsymbol{B}$ then vanishes, $\boldsymbol{J}\cdot\boldsymbol{B} = 0$. If, however, the random motions are helical, each local loop has an additional systematic twist, as shown in panel (b) for the half-space $z > 0$ assuming that the angular velocity of the overall rotation is parallel to the z-axis (compare with Fig. 7.2).

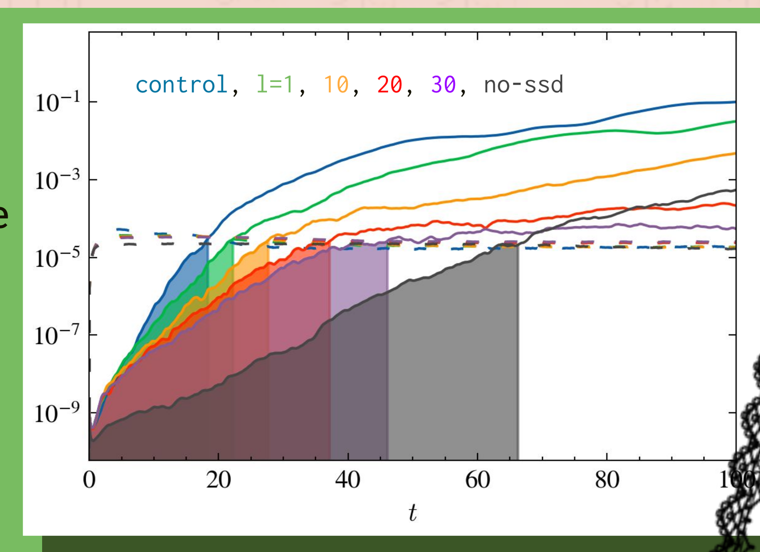
Excerpt from “Astrophysical Magnetic Fields” - Shukurov & Subramanian, describing why helical fields lead to a mean-field dynamo. The QR contains the DOI!



Modulating the kinetic helicity of the flow helps one control little “bubbles” of α, corresponding to local mean-field dynamos, each contributing to the overall mean magnetic field growth.

As one decreases the size of the bubbles, the growth rate of the mean-field dynamo reduces.

A mean-field dynamo **DOES** operate.



THE $$$ PLOT

— $|\langle \boldsymbol{B} \rangle|^2$ - - $0.002u_{\mathrm{rms}}^2$